Tutorial 10: Support Vector Machines

Margin geometry, hard and soft margins, and the dual

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Task 1: Comparing linear separators

Four labeled two-feature observations are:

i	$\boldsymbol{x}_i = (x_{i1}, x_{i2})^\mathsf{T}$	y_i
1	$(3, 1)^\mathsf{T}$	$+1$
2	$(2, 2)^\mathsf{T}$	$+1$
3	$(0, 1)^\mathsf{T}$	-1
4	$(1, -1)^\mathsf{T}$	-1

Compare:

$$f_A(\boldsymbol{x}) = x_1 - \frac{3}{2}, \qquad f_B(\boldsymbol{x}) = 2x_1 + x_2 - \frac{7}{2}.$$

For each, $\boldsymbol{w}$ is the coefficient vector and b is the intercept.

(a) Compute all scores and verify the predicted classes.

(b) Compute $\gamma_{\min} = \min_i y_i f(\boldsymbol{x}_i)/\|\boldsymbol{w}\|_2$ for both separators. Which has the larger margin?

(c) Rescale f_B canonically, then give its two margin lines and full margin width.

(d) Identify the margin-contact points. Verify that scaling f_B by 4 preserves the geometric margin of observation 2.

Task 2: Hard-margin optimality

A two-dimensional training set has the following classes:

$$\begin{array}{l|l} y=-1 & (0,0),\ (-1,0),\ (0,-1) \\ y=+1 & (2,2),\ (3,2),\ (2,3) \end{array}$$

All coordinates denote column vectors in $\mathbb{R}^2$. Use $f(\boldsymbol{x}) = \boldsymbol{w}^\top \boldsymbol{x} + b$ and the constraints $y_i f(\boldsymbol{x}_i) \geq 1$.

(a) The closest opposite-class pair is $(0,0)$ and $(2,2)$. Use their midpoint and connecting direction to find a canonically scaled candidate $\boldsymbol{w}, b$.

(b) Compute signed scores for all six observations and identify the margin-contact points.

(c) Compute the one-sided and full geometric margins and the primal objective $\|\boldsymbol{w}\|_2^2/2$.

(d) Prove the candidate is optimal by subtracting the two closest-point constraints and applying Cauchy–Schwarz.

Task 3: Soft-margin trade-off

For one-dimensional observations, compare two candidate classifiers:

x_i	-2	-0.5	0.2	2
y_i	-1	-1	$+1$	$+1$

$$f_A(x) = x + 0.1, \qquad f_B(x) = 0.5x + 0.5.$$

Use signed scores $m_i = y_i f(x_i)$, minimum slack $\xi_i = \max(0, 1 - m_i)$, and $J = \frac{1}{2}w^2 + C\sum_i \xi_i$, where $C > 0$.

(a) Compute the four signed scores and minimum slacks for both candidates.

(b) Express $J_A(C)$ and $J_B(C)$ exactly.

(c) Compare the candidates at $C = 0.2$ and $C = 1$.

(d) Find the value of C at which their objectives are equal. Explain the trade-off, including the error made by one candidate.

Task 4: Dual and support vectors

Consider three two-dimensional training observations:

i	$\boldsymbol{x}_i$	y_i
1	$(-1, 0)^\mathsf{T}$	-1
2	$(0, -1)^\mathsf{T}$	-1
3	$(1, 1)^\mathsf{T}$	$+1$

The hard-margin dual maximizes $W(\boldsymbol{\alpha}) = \sum_i \alpha_i - \frac{1}{2}\| \sum_i \alpha_i y_i \boldsymbol{x}_i \|_2^2$, with $\alpha_i \geq 0$ and $\sum_i \alpha_i y_i = 0$.

(a) Set $a = \alpha_1$, $c = \alpha_2$. Use the equality constraint to express α_3 and the primal weight $\boldsymbol{w}$ in terms of a, c.

(b) Reduce W to two variables and find all optimal multipliers.

(c) Recover $\boldsymbol{w}, b$, verify all training margins, and identify the support vectors.

(d) Compute the full margin width and classify $(1, 0)^\mathsf{T}$ and $(0, 0)^\mathsf{T}$.

Tutorial 10 complete

Questions and discussion